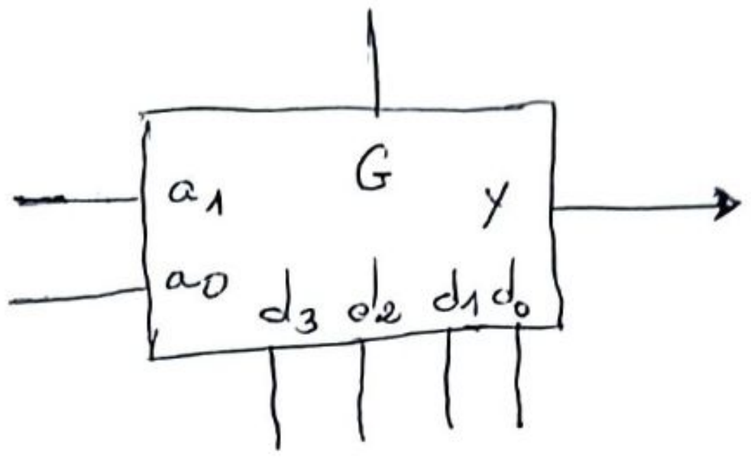
Exercice 1 : (10 pts)

1/

*)

G	a_1	a_0	Y
1	x	x	1
0	0	0	d_0
0	0	1	d_1
0	1	0	d_2
0	1	1	d_3

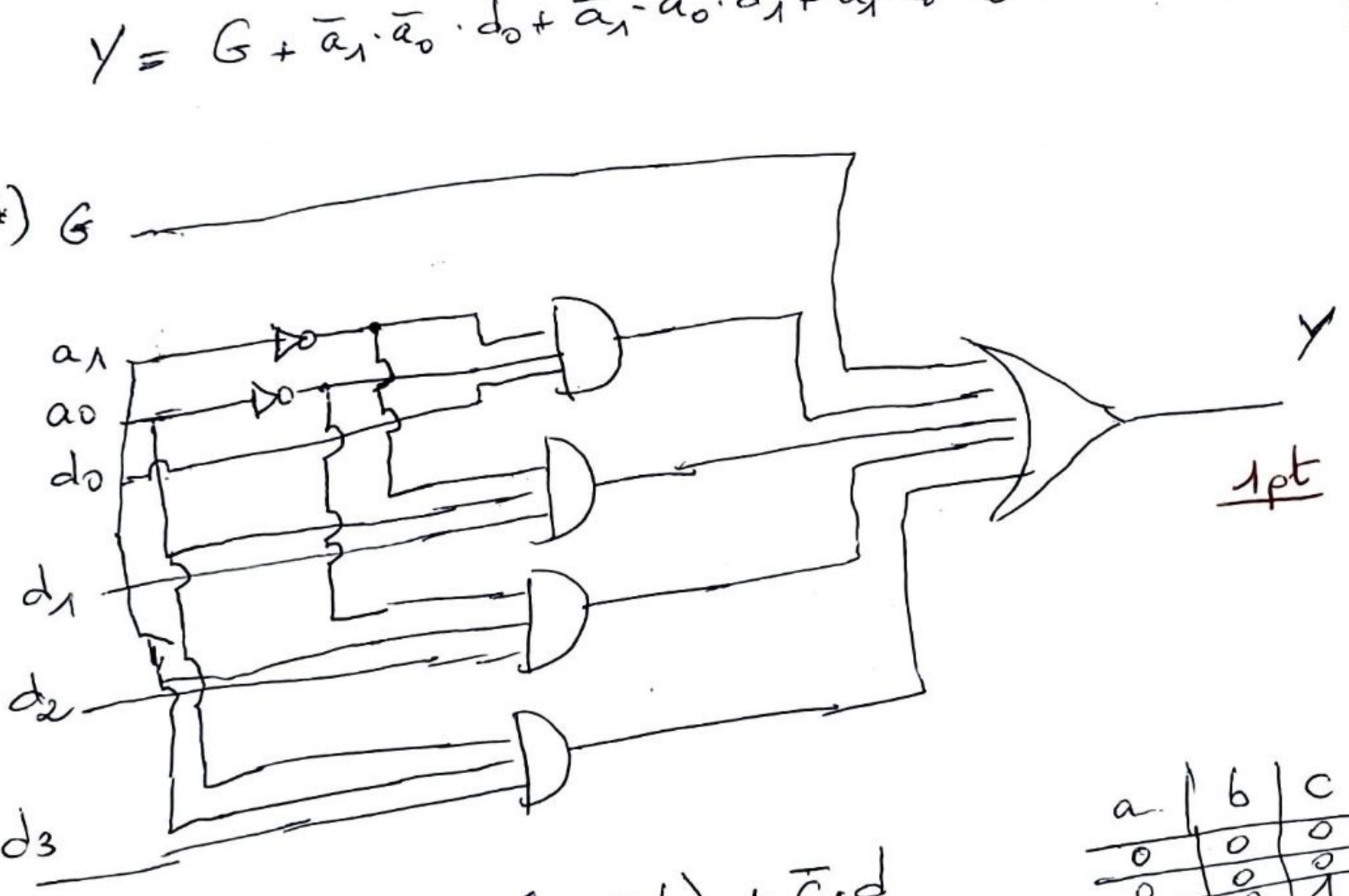
1 pt



*) $Y = G + \bar{G} \cdot \left(\bar{a_1} \cdot \bar{a_0} \cdot d_0 + \bar{a_1} \cdot a_0 \cdot d_1 + a_1 \cdot \bar{a_0} \cdot d_2 + a_1 \cdot a_0 \cdot d_3 \right)$

$Y = G + \bar{a_1} \cdot \bar{a_0} \cdot d_0 + \bar{a_1} \cdot a_0 \cdot d_1 + a_1 \cdot \bar{a_0} \cdot d_2 + a_1 \cdot a_0 \cdot d_3$ 1 pt

*)



1 pt

2/ $F(a,b,c,d) = (a \oplus b) + \bar{c} \cdot d$

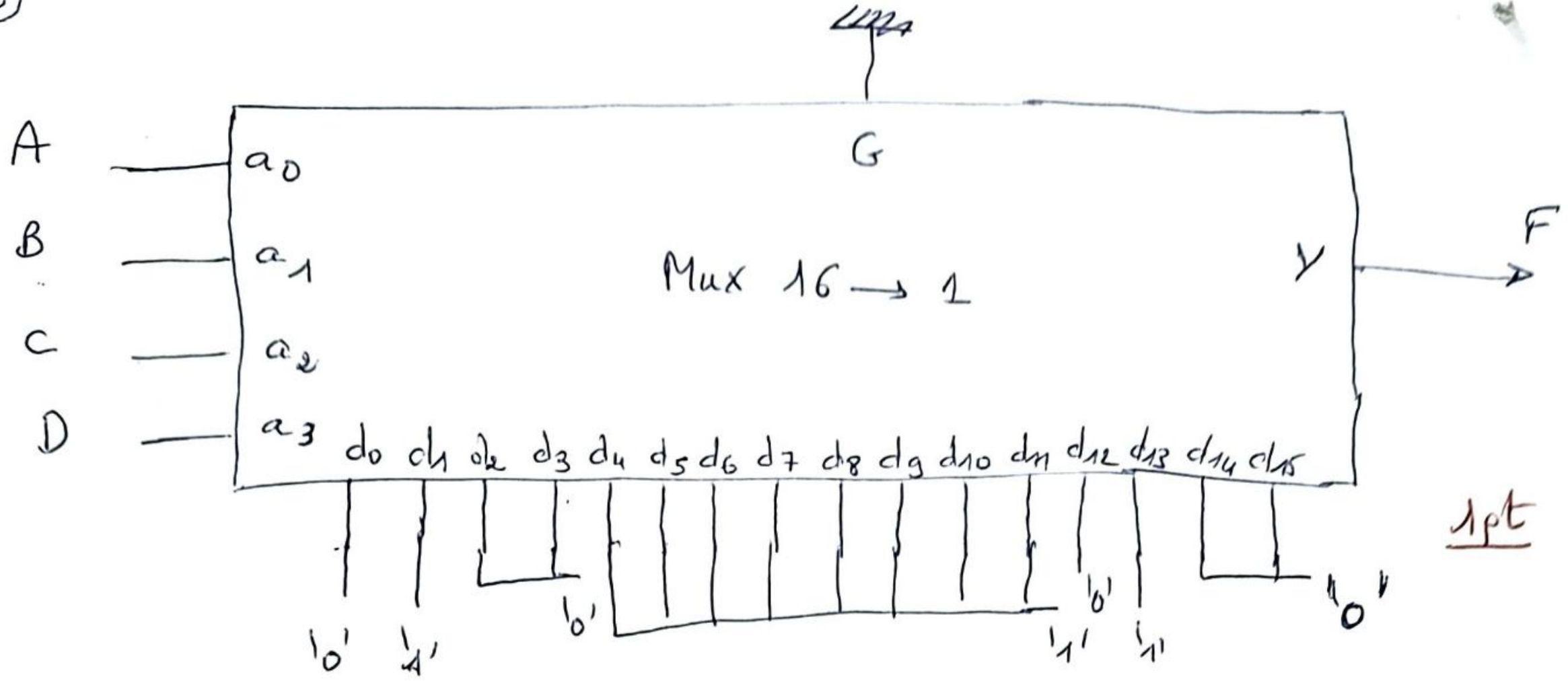
$= \bar{a} \cdot b + a \cdot \bar{b} + \bar{c} \cdot d$

2 pts

a	b	c	d	F	
0	0	0	0	0	
0	0	0	1	1	↑
0	0	1	0	0	
0	0	1	1	0	
0	1	0	0	1	4
0	1	0	1	1	5
0	1	1	0	1	6
0	1	1	1	1	7
1	0	0	0	1	8
1	0	0	1	1	9
1	0	1	0	1	10
1	0	1	1	1	11
1	1	0	0	0	
1	1	0	1	1	13
1	1	1	0	0	
1	1	1	1	0	

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(a)



1 pt

(b) A l'aide d'un Mux 8 → 1

$F(a,b,c,d) = \bar{a}.b + a.\bar{b} + \bar{c}.d$

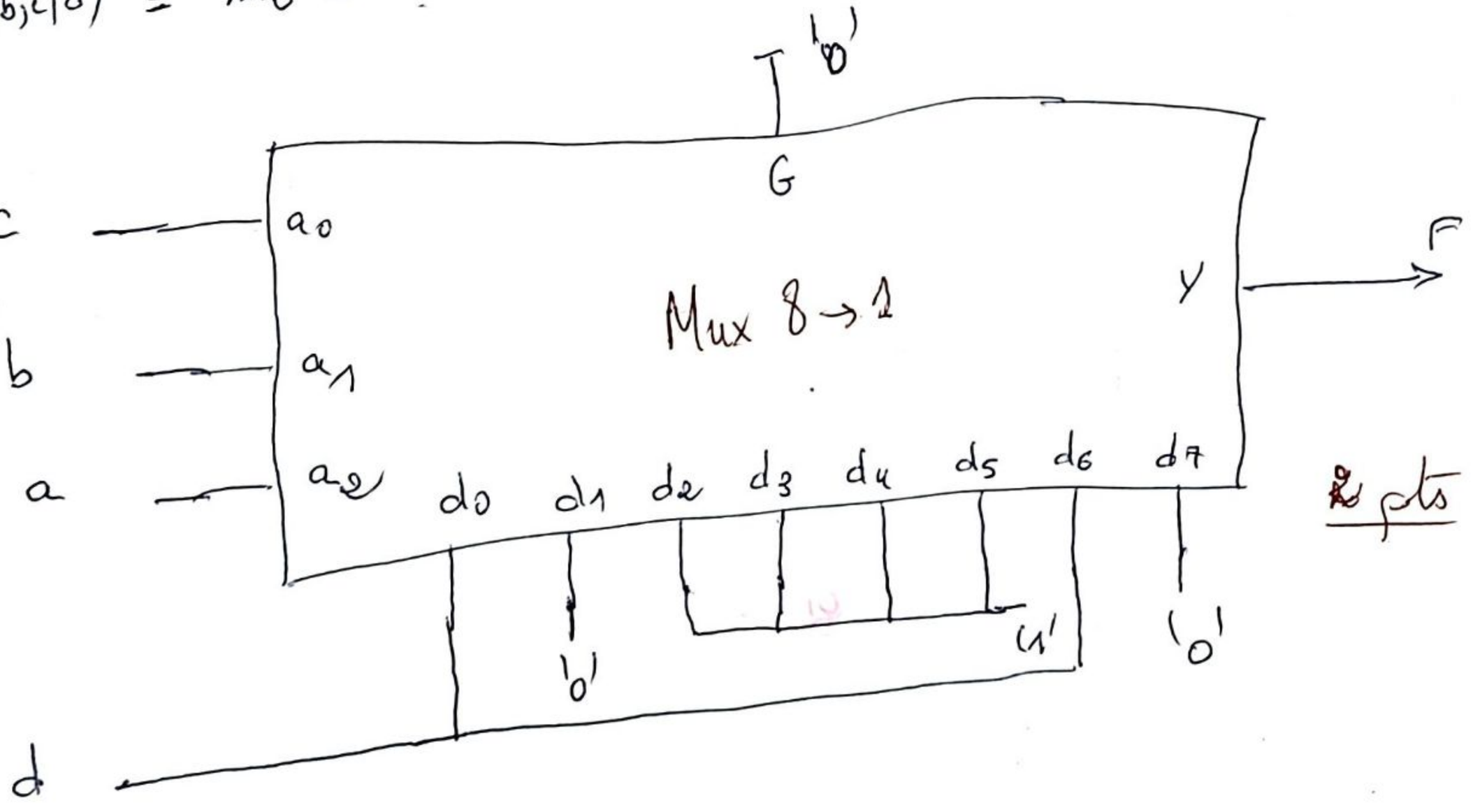
$= \bar{a}.b.(\bar{c}+c) + a.\bar{b}.(c+\bar{c}) + (a+\bar{a}).(b+\bar{b}).\bar{c}.d$

$= \bar{a}.b.\bar{c} + \bar{a}.b.c + a.\bar{b}.\bar{c} + a.\bar{b}.c + a.b.\bar{c}.d + a.\bar{b}.\bar{c}.d + \bar{a}.b.\bar{c}.d + \bar{a}.\bar{b}.\bar{c}.d$

$= \bar{a}.\bar{b}.\bar{c}.d + \bar{a}.b.\bar{c}.(\overbrace{1+d}^{=1}) + \bar{a}.b.c + a.\bar{b}.\bar{c}.(1+d) + a.\bar{b}.c + a.b.\bar{c}.d$

2 pts

$F(a,b,c,d) = m_0.d + m_2 + m_3 + m_4 + m_5 + m_6.d$



2 pts

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Exercice 2 :

Compteur asynchrone modulo 10 (0 → 9)

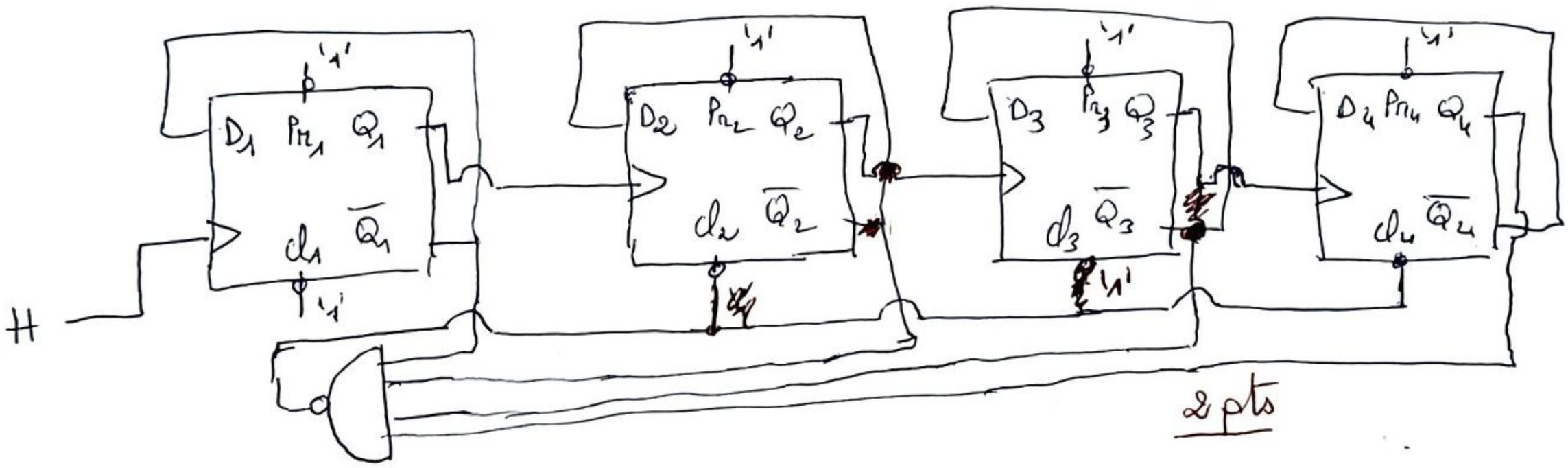
4 bascules

Q_4	Q_3	Q_2	Q_1
0	0	0	0
0	0	0	1
⋮	⋮	⋮	⋮
1	0	0	1
1	0	1	0 ← Moment fugitif → Forçage à '0' de Q_4 et Q_2
0	0	0	0

→ cl_4 et cl_2

1 pt

$cl_4 = cl_2 = \overline{Q_4 \cdot \bar{Q}_3 \cdot \bar{Q}_2 \cdot \bar{Q}_1}$ 1 pt



2 pts

Exercice 3 : (6 pts)

1/ **1ère bascule**

$J_A = 1$

$K_A = 1$

2ème bascule

$J_B = Q_A$

$K_B = Q_A \cdot Q_C$

3ème bascule

$J_C = Q_A \cdot Q_B$

$K_C = Q_A$

1,5 pts

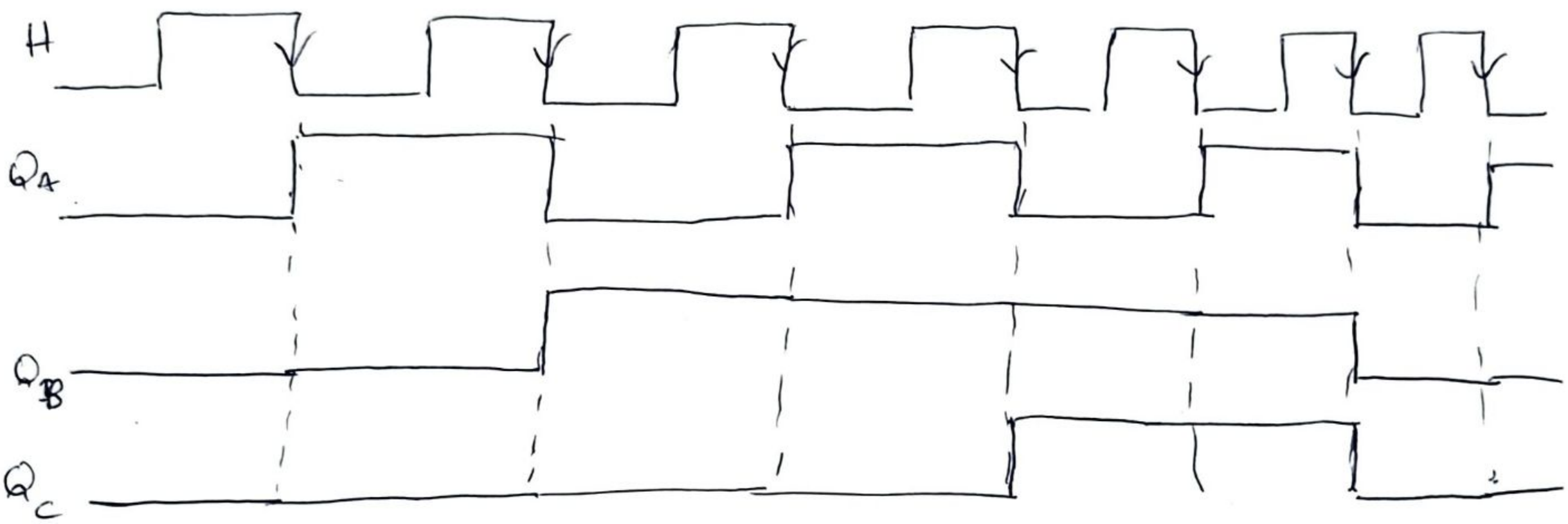
Table de vérité de la bascule JK

J	K	Q
0	0	Q^-
0	1	0
1	0	1
1	1	$\overline{Q^-}$

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Q_C	Q_B	Q_A	J_C	K_C	J_B	K_B	J_A	K_A
0	0	0	0	0	0	0	1	1
0	0	1	0	1	1	0	1	1
0	1	0	0	0	0	0	1	1
0	1	1	1	1	1	0	1	1
1	1	0	0	0	0	0	1	1
1	1	1	1	1	1	1	1	1
0	0	0						

3 pts



1,5 pts

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